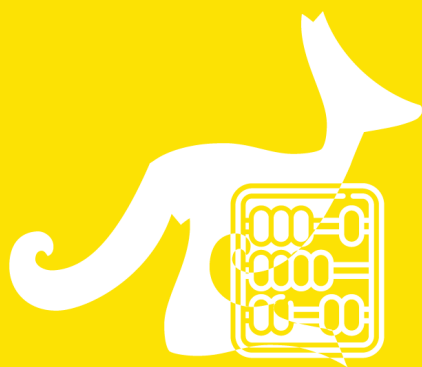




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a

2B pencil.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

Good Luck!

预祝考试顺利

**Year 9
&
Year 10**

3 points

1. Each year, the third Thursday in March is named Kangaroo Day. The dates of Kangaroo Day for the next few years are shown below, with one error. Which date is wrong?

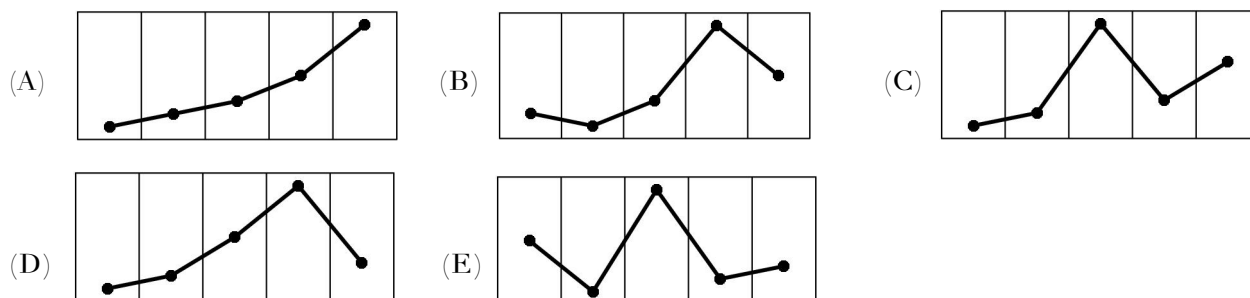
每年三月份的第三个星期四是“袋鼠日”。接下来几年的“袋鼠日”的日期如下所示，其中有一个日期是错误的。请问哪一个日期是错误的？

- (A) 2022 17 March 2022 年 3 月 17 日 (B) 2023 16 March 2023 年 3 月 16 日
(C) 2024 14 March 2024 年 3 月 14 日 (D) 2025 20 March 2025 年 3 月 20 日
(E) 2026 19 March 2026 年 3 月 19 日

2. Jenny looks at her weather app that shows the predicted weather and maximum temperatures for the next five days. Which of the following represents the corresponding graph of maximum temperatures?

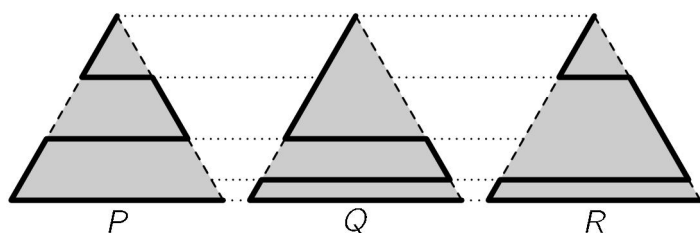
Jenny 的天气应用软件上显示了未来 5 天的天气状况和每日最高温度。以下哪个选项可以表示最高温度的变化图像？

-1°C	-2°C	0°C	6°C	2°C
Fri	Sat	Sun	Mon	Tue



3. A park is shaped like an equilateral triangle. A cat wants to walk along one of the three indicated paths (thicker lines) from the upper corner to the lower right corner. The lengths of the paths are P , Q and R , as shown. Which of the following statements about the lengths of the pathes is true?

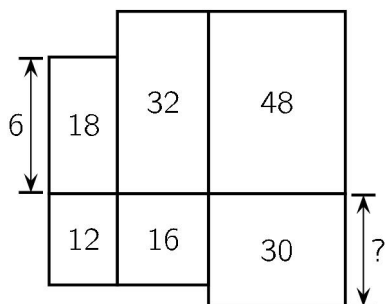
某公园的形状如同一个等边三角形。一只小猫想要沿着三条路线（加粗的线）中的其中一条，从顶端走到右下角。路线的长度分别为 P , Q , R （如图所示）。以下关于路线长度的表述中，哪一项是正确的？



- (A) $P < Q < R$ (B) $P < R < Q$ (C) $P < Q = R$ (D) $P = R < Q$ (E) $P = Q = R$

4. Six rectangles are arranged as shown. The top left-hand rectangle has height 6 cm. The numbers within the rectangles indicate their areas in cm^2 . What is the height of the bottom right-hand rectangle?

六个长方形的排列方式如下。左上角长方形的高度为 6 cm，长方形内的数代表它的面积（单位为 cm^2 ）。请问右下角长方形的高度是多少？



- (A) 4 cm (B) 5 cm (C) 6 cm (D) 7.5 cm (E) 10 cm

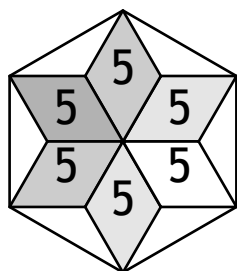
5. The halftime score of a handball match was 9 : 14, thus the visiting team was leading by five goals. As a consequence of coach instructions received at halftime, the home team dominated in the second half and scored twice as many goals as their opponents. The home team won the match by one goal. What was the final score of the match?

在一场手球比赛中，上半场的比分是 9 : 14，客队领先 5 个球。由于在中场休息时收到教练的指示，主队在下半场占据了上风，进球数是客队的两倍。最终主队以一球之差赢得了比赛。请问这场比赛的最终比分是多少？

- (A) 20 : 19 (B) 21 : 20 (C) 22 : 21 (D) 23 : 22 (E) 24 : 23

6. Six congruent rhombuses, each of area 5 cm^2 , form a star. The tips of the star are joined to draw a regular hexagon, as shown. What is the area of the hexagon?

如图所示，六个面积均为 5 cm^2 的全等菱形构成一个星形，星形的顶点连接在一起构成一个正六边形。请问这个正六边形的面积是多少？



- (A) 36 cm^2 (B) 40 cm^2 (C) 45 cm^2 (D) 48 cm^2 (E) 60 cm^2

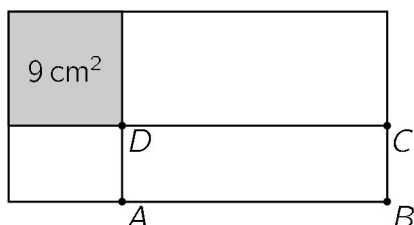
7. In a jazz band, Giuseppe plays the saxophone, Sergio plays the trumpet, and Eliana sings. They are all the same age. There are three more members of the jazz band, who are 19, 20 and 21 years old respectively. The average age of the jazz band is 21. How old is Eliana?

在一个爵士乐队中，Giuseppe 吹萨克斯，Sergio 吹小号，Eliana 唱歌。他们三个人的年龄相同。乐队里还有另外 3 名成员，年龄分别是 19 岁、20 岁和 21 岁。已知乐队成员的平均年龄是 21 岁，请问 Eliana 的年龄是多少岁？

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

8. A rectangle with perimeter 30 cm is divided into four parts by a vertical line and a horizontal line. One of the parts is a square of area 9 cm^2 , as shown in the figure. What is the perimeter of rectangle ABCD?

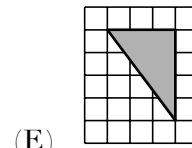
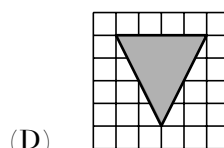
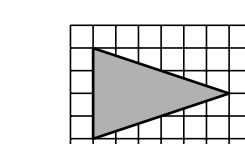
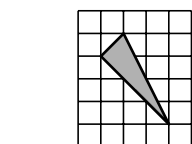
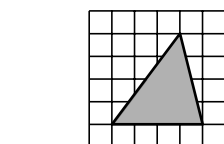
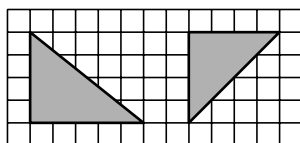
如图所示，一条垂直线和一条水平线将一个周长为 30 cm 的长方形分成了 4 个区域。其中一个区域是面积为 9 cm^2 的正方形。请问长方形 ABCD 的周长是多少？



- (A) 14 cm (B) 16 cm (C) 18 cm (D) 21 cm (E) 24 cm

9. Ally drew three triangles on a grid. Exactly two of them have the same area, exactly two of them are isosceles, and exactly two are right-angled triangles. Two of the triangles are shown. Which could be the third one?

Ally 在网格图中画了 3 个三角形。在这 3 个三角形中，恰好有两个面积相等，有两个是等腰三角形，有两个是直角三角形。其中两个三角形如下图所示。请问下列哪个选项可能是第三个三角形的形状？



10. The little kangaroo has chosen a special number. She gets the same result when she subtracts $\frac{1}{10}$ from her number as she does when she multiplies it by $\frac{1}{10}$. What is her number?

小袋鼠选择了一个特别的数，这个数减 $\frac{1}{10}$ 得到的结果和该数乘以 $\frac{1}{10}$ 得到的结果相同。请问她选择的这个数是多少？

(A) $\frac{1}{100}$

(B) $\frac{1}{11}$

(C) $\frac{1}{10}$

(D) $\frac{11}{100}$

(E) $\frac{1}{9}$

4 points

11. Tom had ten sparklers of the same size. He lit one first. When only a tenth of it remained, he lit the second one. When only a tenth of that remained, he lit the third one, and so on. Sparklers burn at the same speed along their entire length. One sparkler will burn in 2 minutes. How long did it take for all 10 sparklers to burn down?

Tom 有 10 个同样大小的烟花棒。他先点燃了一个，在烟花燃烧的只剩下十分之一的时候，他又点燃了第二个。在第二个烟花燃烧的只剩下十分之一的时候又点燃了第三个，以此类推。烟花在整个燃烧过程的速度保持不变，并且每个烟花均在两分钟内烧完。请问这 10 个烟花棒全部烧完需要多长时间？

(A) 18 min 20 sec 18 分 20 秒

(B) 18 min 12 sec 18 分 12 秒

(C) 18 min 18 分钟

(D) 17 min 17 分钟

(E) 16 min 40 sec 16 分 40 秒

12. Ahmad walks up 8 steps going up either 1 or 2 steps at a time. There is a hole on the 6th step, so he cannot use this step. In how many different ways can Ahmad reach the top step?

Ahmad 要走上 8 个台阶，每次走 1 个或 2 个台阶。第六个台阶上有一个洞，所以他不能走这个台阶。请问如果要走到最高的台阶，Ahmad 有多少种不同的走法？

(A) 6

(B) 7

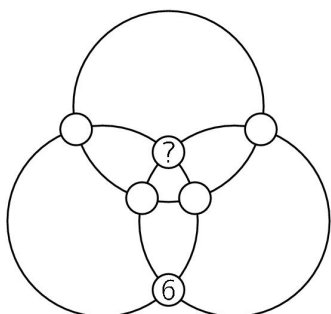
(C) 8

(D) 9

(E) 10

13. The numbers from 1 to 6 are placed in the circles at the intersections of three rings. The position of number 6 is shown. The sums of the numbers on each ring are the same. What number is placed in the circle with the question mark?

将数字 1 到 6 放在三个圆环相交的圆圈里。数字 6 的位置如下图所示，每个圆环上的数字之和是相同的。请问标有问号的圆圈里的数字是多少？



(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

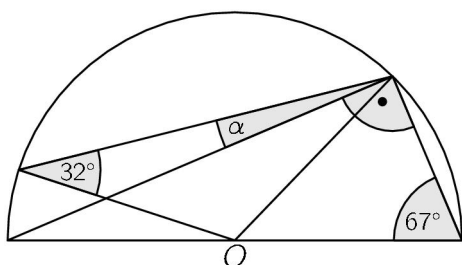
14. 2021 has a remainder of 5 when divided by 6, by 7, by 8, and by 9. How many positive integers, less than 2021, have this property?

2021 除以 6、7、8、9 余数都是 5。请问小于 2021 的正整数中，有多少个数具有同样的性质？

- (A) 4 (B) 3 (C) 2 (D) 1 (E) none 以上都不是

15. The figure shows a semicircle with center O . Two of the angles are given. What is the size, in degrees, of the angle α ?

下图显示了一个以 O 为圆心的半圆，已知其中两个角的度数。请问角 α 的度数是多少？



- (A) 9° (B) 11° (C) 16° (D) 17.5° (E) 18°

16. In a team competition, there are five teams waiting to start. Each team consists of either only boys or only girls. The number of team members are 9, 15, 17, 19 and 21. After all members of the first team have started, the number of girls not started yet is three times the number of boys not started yet. How many members are on the team that has already started?

在一场团队比赛中，一共有 5 支队伍正在等待比赛开始。每支队伍里要么只有男生，要么只有女生。每个队伍人数为 9 人、15 人、17 人、19 人、21 人。第一支队伍中所有成员都开始比赛后，尚未开始比赛的女生人数是尚未开始比赛的男生人数的三倍。请问已经开始比赛的队伍中有多少名成员？

- (A) 9 (B) 15 (C) 17 (D) 19 (E) 21

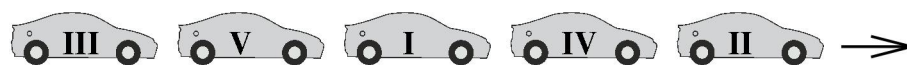
17. Five cars participated in a race, starting in the order shown.

五辆汽车进行比赛，它们的出发顺序如下。



Whenever a car overtook another car, a point was awarded. The cars reached the finish line in the following order:

一辆汽车无论何时超过另一辆汽车均可得 1 分。汽车到达终点的顺序如下：



What is the smallest number of points in total that could have been awarded?

请问所有车辆总得分的最小可能值是多少？

- (A) 10 (B) 9 (C) 8 (D) 7 (E) 6

18. A 3×3 square initially has the number 0 in each of its cells. In one step all four numbers in one 2×2 subsquare such as the shaded one, for example, are then increased by 1. This operation is repeated several times to obtain the arrangement on the right. Unfortunately some numbers in this arrangement are hidden. What number is in the square with the question mark?

一个 3×3 的正方形网格中每个单元格中刚开始都写有数字 0，之后每变化一次，其中 2×2 的小正方形中（比如灰色部分）的 4 个数字都增加 1。该过程重复多次，最终得到第二张图片的形状，但有些数字被覆盖了。请问带问号的方框中的数字是多少？

0	0	0
0	0	0
0	0	0

	18	
	47	
13		?

(A) 14

(B) 15

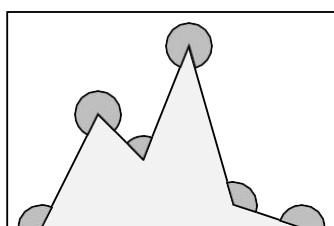
(C) 16

(D) 17

(E) 19

19. What is the sum of the six marked angles in the picture?

在这幅图中，标记出来的六个角的度数之和是多少？



(A) 360°

(B) 900°

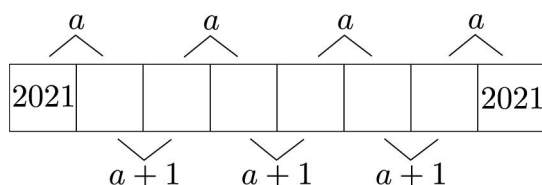
(C) 1080°

(D) 1120°

(E) 1440°

20. There are eight boxes in the strip shown. Numbers in adjacent boxes have sum a or $a + 1$ as shown. The numbers in the first box and the eighth box are both 2021. What is the value of a ?

如图所示，图中有 8 个盒子，相邻盒子的数字之和是 a 或 $a + 1$ 。已知第一个盒子和第八个盒子中的数都是 2021。请问 a 的值是多少？



(A) 4041

(B) 4042

(C) 4043

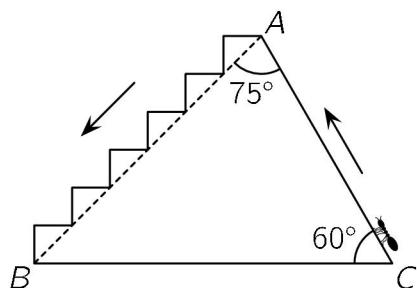
(D) 4044

(E) 4045

5 points

21. An ant climbs from C to A on path CA and descends from A to B on the stairs, as shown in the diagram. What is the ratio of the lengths of the ascending and descending paths?

如图所示，一只蚂蚁沿路线 CA 从 C 爬到 A ，然后顺着阶梯从 A 爬到 B 。请问往上爬的路径长度和往下爬的路径长度之比是多少？



- (A) 1 (B) $1/2$ (C) $1/3$ (D) $\sqrt{2}/2$ (E) $\sqrt{3}/3$

22. The numbers a, b and c satisfy $a + b + c = 0$ and $abc = 78$. What is the value of $(a+b)(b+c)(c+a)$?

数字 a, b 和 c 满足 $a + b + c = 0$ ，且 $abc = 78$ 。请问 $(a+b)(b+c)(c+a)$ 的值是多少？

- (A) -156 (B) -39 (C) 78 (D) 156
(E) none of the previous 以上选项都不正确

23. Let N be the smallest positive integer whose sum of its digits is 2021. What is the sum of the digits of $N + 2021$?

设 N 为满足各个数位上的数字之和为 2021 的最小正整数。请问 $N + 2021$ 的各个数位上的数字之和是多少？

- (A) 10 (B) 12 (C) 19 (D) 28 (E) 2021

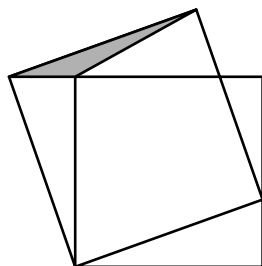
24. Three boys played a “Word” game in which they each wrote down 10 words. Each boy scored three points if neither of the other boys had the same word. Each boy scored one point if only one of the other boys had the same word. No points were awarded for words which all three boys had. When they added up their scores, they found that they each had different score. Sam had 19 points, which was the smallest score, and James had the highest score. How many points did James score?

三个男孩玩“单词”游戏，每个男孩都写下 10 个单词。如果一个男孩写出的单词和其他两个男孩都不一样，那么这个男孩可以得 3 分；如果一个男孩写出的单词只和其中一个男孩有相同的单词，那么这个男孩可以得 1 分；如果一个男孩写出的单词和其他两个男孩都一样，则这个男孩不得分。当他们三个人把各自的得分加总后，发现三人的分数都不一样。Sam 得分最低，得了 19 分，James 得分最高。请问 James 得了多少分？

- (A) 20 (B) 21 (C) 23 (D) 24 (E) 25

25. The smaller square in the picture has area 16 and the grey triangle has area 1. What is the area of the larger square?

图中较小的正方形面积为 16，灰色三角形的面积为 1。请问较大的正方形面积是多少？



- (A) 17 (B) 18 (C) 19 (D) 20 (E) 21

26. Each of the numbers a and b is a square of an integer. The difference $a - b$ is a prime number. Which of the following could be b ?

数 a 和数 b 都是一个整数的平方。 $a - b$ 的值是一个质数。以下选项中，哪一个可能是 b 的值？

- (A) 100 (B) 144 (C) 256 (D) 900 (E) 10000

27. In the 4×4 table some cells must be painted black. The numbers next to and below the table show how many cells in that row or column must be black. In how many ways can this table be painted?

在一个 4×4 的表格中，一些单元格必须涂成黑色。表格旁边和下面的数字表示该行或该列黑色单元格的数目。请问这个表格有多少种填涂方式？

				2
				0
				2
				1
2	0	2	1	

- (A) 1 (B) 2 (C) 3 (D) 5 (E) more than 5 多于 5 种

28. How many five-digit positive numbers have the product of their digits equal to 1000?

有多少个五位数是正整数，且各个数位上的数字乘积等于 1000？

- (A) 10 (B) 20 (C) 30 (D) 40 (E) 60

29. Christina has eight coins whose weights in grams are different positive integers. When Christina puts any two coins on one side of a balance scales and any two on the other side of the balance scales, the side containing the heaviest of the four coins is always the heavier side. What is the smallest possible weight of the heaviest coin?

Christina 有 8 枚硬币，每一枚硬币的重量以克计算，都是不同的正整数。当 Christina 把任意两枚硬币放在天平的一端，把另外两枚硬币放在天平的另一端时发现，放有这四枚硬币中最重的硬币的那一端总是较重的一端。请问最重的那枚硬币重量的最小可能值是多少克？

- (A) 8 (B) 12 (C) 34 (D) 128 (E) 256

30. 2021 balls are arranged in a row and are numbered from 1 to 2021. Each ball is coloured in one of four colours: green, red, yellow or blue. Among any five consecutive balls there is exactly one red, one yellow and one blue ball. After any red ball the next ball is yellow. The balls numbered 2, 20 and 202 are green. What colour is the ball numbered 2021?

把 2021 个球排成一行，依次标记为 1 到 2021。每个球的颜色都是绿色、红色、黄色和蓝色中的一种。在任意连续五个球中，总是有一个红球，一个黄球和一个蓝球。红球后面一定是黄球。已知第 2 个、第 20 个和第 202 个球都是绿球，请问第 2021 个球是什么颜色？

- (A) Green 绿色 (B) Red 红色 (C) Yellow 黄色
(D) Blue 蓝色 (E) It is impossible to determine. 无法判断

PROBLEMS WITH SOLUTIONS

3 points

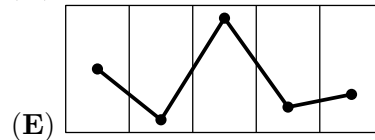
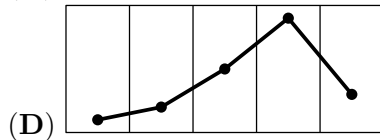
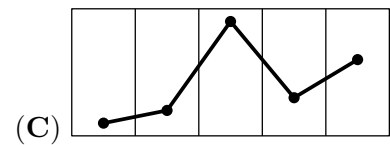
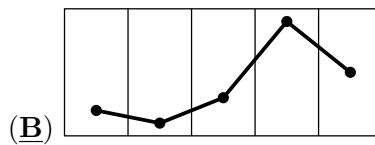
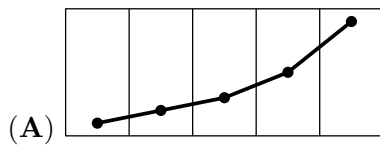
1. Each year, the third Thursday in March is named Kangaroo Day. The dates of Kangaroo Day for the next few years are shown below, with one error. Which date is wrong?

(A) 2022 17 March (B) 2023 16 March (C) 2024 14 March (D) 2025 20 March (E) 2026 19 March

SOLUTION: (C) is wrong because it's the second Thursday of March 2024.

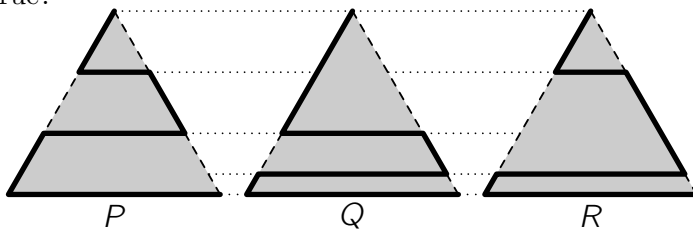
2. Jenny looks at her weather app that shows the predicted weather and maximum temperatures for the next five days. Which of the following represents the corresponding graph of maximum temperatures?

-1°C	-2°C	0°C	6°C	2°C
Fri	Sat	Sun	Mon	Tue



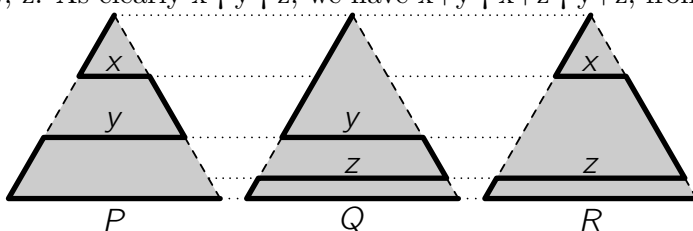
SOLUTION: 4th day is the warmest, only (B) or (D) could be right. 2nd day is the coldest, (B) is the solution.

3. A park is shaped like an equilateral triangle. A cat wants to walk along one of the three indicated paths (thicker lines) from the upper corner to the lower right corner. The lengths of the paths are P, Q and R, as shown. Which of the following statements about the lengths of the pathes is true?

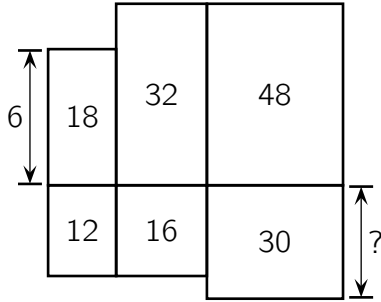


(A) $P < Q < R$ (B) $P < R < Q$ (C) $P < Q = R$ (D) $P = R < Q$ (E) $P = Q = R$

SOLUTION: The parts of the path on the edges of the equilateral triangles are equal in all cases, so we only need to compare the internal part of the paths. Let the internal parts of the path have lengths x , y , z . As clearly $x \leq y \leq z$, we have $x+y \leq x+z \leq y+z$, from which it follows $P \leq R \leq Q$.



4. Six rectangles are arranged as shown. The top left-hand rectangle has height 6 cm. The numbers within the rectangles indicate their areas in cm^2 . What is the height of the bottom right-hand rectangle?



- (A) 4 cm (B) 5 cm (C) 6 cm (D) 7.5 cm (E) 10 cm

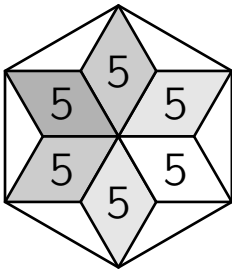
SOLUTION: The width of the top left rectangle is $18:6=3$ cm, so the bottom left rectangle also has width 3 cm so its height is $12:3=4$ cm. Now we look at the second column and work in a similar manner starting from the bottom rectangle. In the end we can calculate all heights and widths.

5. The halftime score of a handball match was 9 : 14, thus the visiting team was leading by five goals. As a consequence of coach instructions received at halftime, the home team dominated in the second half and scored twice as many goals as their opponents. The home team won the match by one goal. What was the final score of the match?

- (A) 20 : 19 (B) 21 : 20 (C) 22 : 21 (D) 23 : 22 (E) 24 : 23

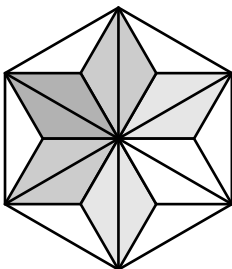
SOLUTION: The home team has 5 goals less at halftime. To be better in the end it needs 6 goals more and so the score of the second half must have been 12 : 6. 21 : 20 is the final score of the match.

6. Six congruent rhombuses, each of area 5 cm^2 , form a star. The tips of the star are joined to draw a regular hexagon, as shown. What is the area of the hexagon?



- (A) 36 cm^2 (B) 40 cm^2 (C) 45 cm^2 (D) 48 cm^2 (E) 60 cm^2

SOLUTION: The hexagon can be divided into 18 congruent triangles.



Each triangle has an area of 2.5 cm^2 , and so the whole area is $18 \cdot 2.5 \text{ cm}^2 = 45 \text{ cm}^2$.

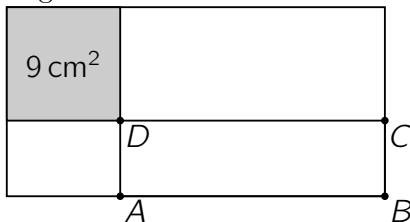
To prove the congruence one can use that each of the original rhombuses has angles of 60, 120, 60 and 120. They are congruent to each other and they all meet in the centre of the circle. The triangles representing half of a rhombus have angles of 30, 30 and 120. So the white triangle have 120 as biggest angle too and they are congruent to them because of SAS (Side-Angle-Side congruence).

7. In a jazz band, Giuseppe plays the saxophone, Sergio plays the trumpet, and Eliana sings. They are all the same age. There are three more members of the jazz band, who are 19, 20 and 21 years old respectively. The average age of the jazz band is 21. How old is Eliana?

- (A) 20 (B) 21 (C) 22 (D) 23 (E) 24

SOLUTION: $3x + 19 + 20 + 21 = 6 \cdot 21 \Leftrightarrow 3x + 60 = 126 \Leftrightarrow 3x = 66 \Leftrightarrow x = 22$

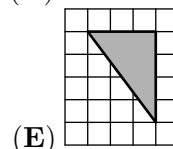
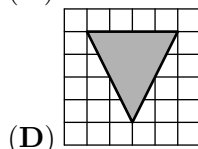
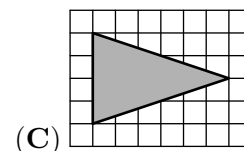
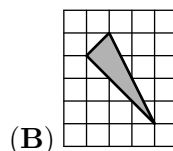
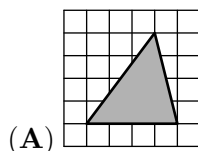
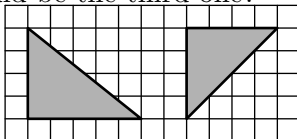
8. A rectangle with perimeter 30 cm is divided into four parts by a vertical line and a horizontal line. One of the parts is a square of area 9 cm^2 , as shown in the figure. What is the perimeter of rectangle ABCD?



- (A) 14 cm (B) 16 cm (C) 18 cm (D) 21 cm (E) 24 cm

SOLUTION: The perimeter of the square plus the perimeter of rectangle ABCD is the perimeter of the original rectangle. Perimeter of the square is 12 cm, so rectangle ABCD has a perimeter of $30 \text{ cm} - 12 \text{ cm} = 18 \text{ cm}$.

9. Ally drew three triangles on a grid. Exactly two of them have the same area, exactly two of them are isosceles, and exactly two are right-angled triangles. Two of the triangles are shown. Which could be the third one?



SOLUTION: Since the two triangles shown are right-angled triangles, their areas are 10 and 8 and only one is isosceles, we are looking for an isosceles, non right-angled triangle of area 8 or 10. Triangle (A) is not isosceles, triangle (B) is too small, triangle (C) has area 12 and triangle (E) is a right-angled triangle. So (D) is the only possible solution as it is an isosceles not right-angled triangle of area 8.

10. The little kangaroo has chosen a special number. She gets the same result when she subtracts $\frac{1}{10}$ from her number as she does when she multiplies it by $\frac{1}{10}$. What is her number?

- (A) $\frac{1}{100}$ (B) $\frac{1}{11}$ (C) $\frac{1}{10}$ (D) $\frac{11}{100}$ (E) $\frac{1}{9}$

SOLUTION: $x - \frac{1}{10} = x \cdot \frac{1}{10} \Leftrightarrow 10x - 1 = x \Leftrightarrow 9x = 1 \Leftrightarrow x = \frac{1}{9}$

4 points

11. Tom had ten sparklers of the same size. He lit one first. When only a tenth of it remained, he lit the second one. When only a tenth of that remained, he lit the third one, and so on. Sparklers burn at the same speed along their entire length. One sparkler will burn in 2 minutes. How long did it take for all 10 sparklers to burn down?

- (A) 18 min 20 sec (B) 18 min 12 sec (C) 18 min (D) 17 min (E) 16 min 40 sec

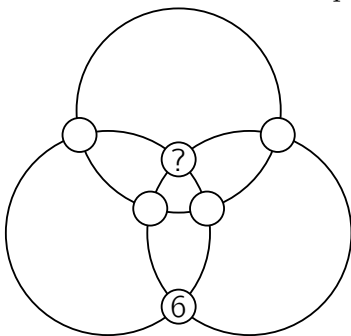
SOLUTION: Each sparkler burns 2 minutes or 120 seconds respectively. 9 times an amount of 12 seconds is saved. 10 times 2 minutes minus 9 times 12 seconds is equal to 9 times 2 minutes plus 12 seconds, (B) is correct.

12. Ahmad walks up 8 steps going up either 1 or 2 steps at a time. There is a hole on the 6th step, so he cannot use this step. In how many different ways can Ahmad reach the top step?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

SOLUTION: There is only one way to move from step 5 to 7 (with 2 steps) and only one way to move from step 7 to 8 (with 1 step). Therefore, the number of different ways to reach step 8 equals the number of ways to reach step 5 in this scenario. The step 5 can be reached in 3 ways with two 2 steps, in 4 ways, with one 2 steps, and in 1 way with zero 2 steps. There are 8 ways in total. Another solution avoiding combinatoric count of all cases: To reach a step (except the first one) we either go one ahead from the previous one or two ahead from the step two steps behind. Total ways to a step (except the first) is the sum of the ways to reach the previous one or the one two steps behind. So the number of steps needed for the first to fifth step are, respectively, 1, then $1 + 1 = 2$, then $1 + 2 = 3$, then $2 + 3 = 5$, then $3 + 5 = 8$ (this is the Fibonacci series).

13. The numbers from 1 to 6 are placed in the circles at the intersections of three rings. The position of number 6 is shown. The sums of the numbers on each ring are the same. What number is placed in the circle with the question mark?



- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

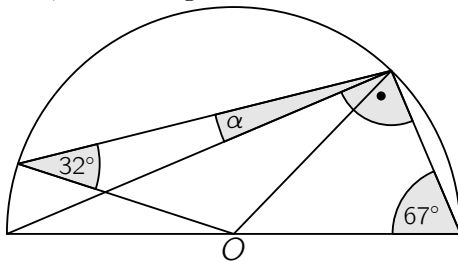
SOLUTION: The sum of the number on each ring is $\frac{2}{3} \cdot (1 + 2 + 3 + 4 + 5 + 6) = 14$. The rings that contain 6 need a further sum of 8 from the other three numbers. This can be done using 1, 2, 5 or 1, 3, 4. The question mark appears in the intersection of two rings containing 6 so must be 1.

14. 2021 has a remainder of 5 when divided by 6, by 7, by 8, and by 9. How many positive integers, less than 2021, have this property?

- (A) 4 (B) 3 (C) 2 (D) 1 (E) none

SOLUTION: Suppose that N is the number with such property. $N - 5$ must be divisible by 6, 7, 8 and 9. The $LCM(6, 7, 8, 9) = 504$ so the possibilities are $0 + 5 = 5$, $504 + 5 = 509$, $1008 + 5 = 1013$ and $1512 + 5 = 1517$.

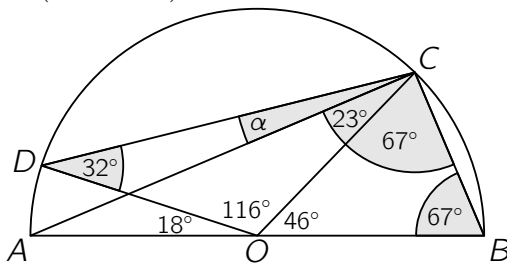
15. The figure shows a semicircle with center O . Two of the angles are given. What is the size, in degrees, of the angle α ?



- (A) 9° (B) 11° (C) 16° (D) 17.5° (E) 18°

SOLUTION: From the isosceles triangles OBC , OCB we find $\angle BOC = 46^\circ$ and $\angle COD = 116^\circ$. So $2\alpha = \angle AOD = 180^\circ - 116^\circ - 46^\circ = 18^\circ$, so $\alpha = 9^\circ$.

second solution: From the isosceles triangle OBC , we find $\angle OCB = 67^\circ$. From the isosceles triangle DOC , we find $\angle DCO = 32^\circ$. Hence $\alpha = \angle DCO - \angle ACO = \angle DCO - (\angle ACB - \angle OCB) = 32^\circ - (90^\circ - 67^\circ) = 9^\circ$.



16. In a team competition, there are five teams waiting to start. Each team consists of either only boys or only girls. The number of team members are 9, 15, 17, 19 and 21. After all members of the first team have started, the number of girls not started yet is three times the number of boys not started yet. How many members are on the team that has already started?

- (A) 9 (B) 15 (C) 17 (D) 19 (E) 21

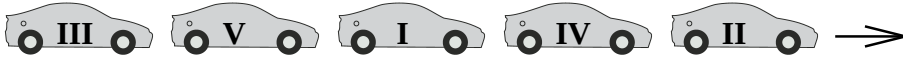
SOLUTION: The total number of players in all teams is equal to 81. As soon as the first team started, there appear to stay three times more girls than boys. The number of players in the first team should have the same remainder by dividing by 4 as 81. There are three numbers 9, 17, 21 with this property. Furthermore one of the remaining teams consists only of boys the other three only of girls and the

number of boys is equal to the mean of the three girls teams. If the first team starting consists of 9 or 17 members, neither of the remaining numbers can be the mean of the other three. Only if the first team consists of 21 members, there will be 15 boys in one remaining team and $9 + 17 + 19 = 45$ girls in the other three teams.

17. Five cars participated in a race, starting in the order shown.



Whenever a car overtook another car, a point was awarded. The cars reached the finish line in the following order:



What is the smallest number of points in total that could have been awarded?

- (A) 10 (B) 9 (C) 8 (D) 7 (E) 6

SOLUTION: For car II to take the lead, it must at least overtake cars III, IV and V (3 points). Car IV must overtake car V since it finished ahead of car V (one point). For I to gain third place, it must at least overtake cars III and V since that end up behind car I (2 points), This gives a minimum of 6 points. The example can be constructed in the similar maner.

18. A 3×3 square initially has the number 0 in each of its cells. In one step all four numbers in one 2×2 subsquare such as the shaded one, for example, are then increased by 1. This operation is repeated several times to obtain the arrangement on the right. Unfortunately some numbers in this arrangement are hidden. What number is in the square with the question mark?

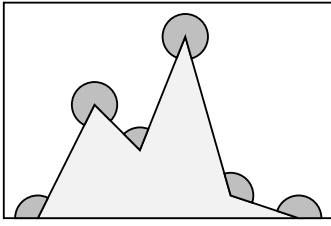
0	0	0		18	
0	0	0		47	
0	0	0	13		?

- (A) 14 (B) 15 (C) 16 (D) 17 (E) 19

SOLUTION: After several such operations we end up with a square as shown. So $d = (a+b+c+d) - (a+b) - c = 47 - 18 - 13 = 16$.

a	$a + b$	b
$a + c$	$a + b + c + d$	$b + d$
c	$c + d$	d

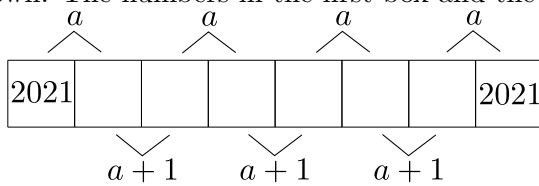
19. What is the sum of the six marked angles in the picture?



- (A) 360° (B) 900° (C) 1080° (D) 1120° (E) 1440°

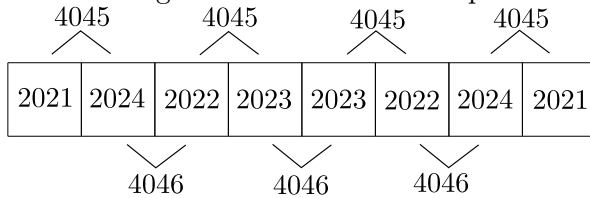
SOLUTION: The six marked angles, together with the four right angles of the rectangular frame make a decagon with interior sum $(10-2) \cdot 180^\circ$. Hence the six marked angles have total $8 \cdot 180^\circ - 4 \cdot 90^\circ = 1080^\circ$.

20. There are eight boxes in the strip shown. Numbers in adjacent boxes have sum a or $a+1$ as shown. The numbers in the first box and the eighth box are both 2021. What is the value of a ?



- (A) 4041 (B) 4042 (C) 4043 (D) 4044 (E) 4045

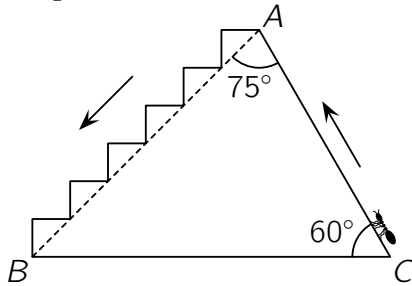
SOLUTION: 2022 should be written on the third position because the sum of this number with the one on second place is 1 more than the sum of the ones in first and second places. Using the same method one can find 2023 on the 5th position and 2024 on the 7th. Thus $a = 4045$. Using the same method from right to left the whole strip should be filled as shown.



A simpler solution is to say that $4a = 3(a+1) + 2021 + 2021 = 3a + 3 + 2021 + 2021 = 3a + 4045$. Hence $a = 4045$.

5 points

21. An ant climbs from C to A on path CA and descends from A to B on the stairs, as shown in the diagram. What is the ratio of the lengths of the ascending and descending paths?



- (A) 1 (B) $1/2$ (C) $1/3$ (D) $\sqrt{2}/2$ (E) $\sqrt{3}/3$

SOLUTION: Draw the altitude AD . As the angle at B is 45° it follows that $AD = BD$. Now, by projecting the zigzag path onto AD , BD we see that its length is $AD + BD = 2AD$. As the

angle at C is 60° it follows that $AD = \frac{1}{2}\sqrt{3}AC \Leftrightarrow AC = \frac{2}{3}\sqrt{3}AD$. So the ratio of the paths is $\frac{AC}{2AD} = \frac{\frac{1}{2}\sqrt{3}AD}{2AD} = \frac{\sqrt{3}}{4}$.

22. The numbers a , b and c satisfy $a+b+c = 0$ and $abc = 78$. What is the value of $(a+b)(b+c)(c+a)$?

- (A) -156 (B) -39 (C) 78 (D) 156
(E) none of the previous

SOLUTION: As $a+b = -c$, $b+c = -a$ and $c+a = -b$, we have $(a+b)(b+c)(c+a) = (-c)(-a)(-b) = -abc = -78$. If you want such an example, take $a = 1.5$, $b = 6.5$ and $c = -8$.

23. Let N be the smallest positive integer whose sum of its digits is 2021. What is the sum of the digits of $N + 2021$?

- (A) 10 (B) 12 (C) 19 (D) 28 (E) 2021

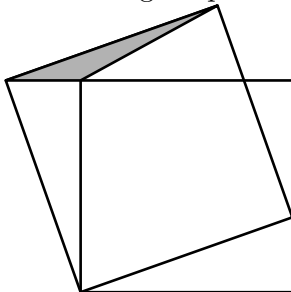
SOLUTION: $N = 59 \dots 9$ (5 followed by 224 times digit 9). Calculate $(N+1)+2020$ gives $60 \dots 02020$. (dots are 218 times 0), therefore $6+2+2=10$.

24. Three boys played a "Word" game in which they each wrote down 10 words. Each boy scored three points if neither of the other boys had the same word. Each boy scored one point if only one of the other boys had the same word. No points were awarded for words which all three boys had. When they added up their scores, they found that they each had different score. Sam had 19 points, which was the smallest score, and James had the highest score. How many points did James score?

- (A) 20 (B) 21 (C) 23 (D) 24 (E) 25

SOLUTION: Words: $19 = 6 \times 3 + 1$. It is not possible: if Sam has 3 times 0, then other 2 boys have as well. But in that case the max point is 21, but 20 is not possible. It means, Sam has $5 \cdot 3 + 4 \cdot 1 + 1 \cdot 0$. Everybody has 1 zero. The 4 times 1 point he shares with the other two: 3 with Noname and 1 with James. The scores are Noname: $6 \cdot 3 + 3 \cdot 1 + 1 \cdot 0 = 21$, James $8 \cdot 3 + 1 \cdot 1 + 1 \cdot 0 = 25$.

25. The smaller square in the picture has area 16 and the grey triangle has area 1. What is the area of the larger square?



- (A) 17 (B) 18 (C) 19 (D) 20 (E) 21

SOLUTION: The length of the bottom side of the triangle and the length of its corresponding height are the same (because of congruence), so it is $\sqrt{2}$. Pythagoras for the left white triangle yields the area of the larger square, $a^2 = \sqrt{2}^2 + 16 = 18$.

26. Each of the numbers a and b is a square of an integer. The difference $a - b$ is a prime number. Which of the following could be b ?

- (A) 100 (B) 144 (C) 256 (D) 900 (E) 10000

SOLUTION: $a = m^2$, $b = n^2$ so $m^2 - n^2 = (m - n)(m + n)$ is a prime, it implies that $m - n = 1$. Then $p = 2m + 1$. The only possibility is $2 \times 30 + 1 = 61$, all other suggested answers lead to composite numbers.

27. In the 4×4 table some cells must be painted black. The numbers next to and below the table show how many cells in that row or column must be black. In how many ways can this table be painted?

				2
				0
				2
				1
2	0	2	1	

- (A) 1 (B) 2 (C) 3 (D) 5 (E) more than 5

SOLUTION: First, it is clear that 5 cells are to be painted because the numbers at the bottom margin add to $2+0+2+1=5$ (we could just as well use the numbers at the right margin where again $2+0+2+1=5$). There are two cases: bottom right corner is painted black or it is not painted black. If the bottom right corner is painted black, then there are no other cells painted in the bottom row and the rightmost column. Thus, the only 4 remaining cells that could be painted are painted. So, there is a unique way to paint the cells if the bottom right corner is painted.

If the bottom right corner is not painted, then one of two cells (first or third column) is painted in the bottom row, and one of two cells (first or third row) is painted in the rightmost column. These two choices are independent, so we have 4 different ways to do this. Once these cells are painted, let's consider the four corners of the 3×3 square formed by removing the bottom row and the rightmost column. Three of the four corners of this square must be painted. The only corner that cannot be painted is the one that is in the intersection of the row and the column of the two cells we already painted. Thus, the other three corners are painted, i.e. the painting is forced (can be completed in a unique way). To summarize, there is 1 way to paint the cells if the bottom right cell is painted and 4 different ways to paint the cells if the bottom right cell is not painted. In total, there are 5 different ways to paint the cells.

An alternative starting with: There are two cases. Suppose the upper left corner is black. Then there are two possibilities for the first row: x-x- or x-x. There are also two possibilities for the first column: x-x- or x-x. Two of them (x-x combined with x-x-) can be completed in a unique way, one (both x-x) can't be completed and one (both x-x-) can be completed in two ways.

If the upper left corner is white, then the table can be filled only in one way. So there are exactly five possibilities.

28. How many five-digit positive numbers have the product of their digits equal to 1000?

- (A) 10 (B) 20 (C) 30 (D) 40 (E) 60

SOLUTION: $1000 = 2^3 \cdot 5^3$ so we must have 3 fives and a pair of digits that multiply up to 8. They are (2, 4) and (1, 8). The cases are symmetric so consider the first one. We can pick the position for the 2 in 5 ways, the position for the 4 in 4 ways. It lead to 20 numbers. The same logic applies to the second case. So the answer is 40.

29. Christina has eight coins whose weights in grams are different positive integers. When Christina puts any two coins on one side of a balance scales and any two on the other side of the balance scales, the side containing the heaviest of the four coins is always the heavier side. What is the smallest possible weight of the heaviest coin?

- (A) 8 (B) 12 (C) 34 (D) 128 (E) 256

SOLUTION: Let $a < b < c < d < e < g < h$ be the weights of the coins. Assume $a = 1$ (otherwise subtract $a - 1$ from each number). Then $b \geq 2$, $c \geq 3$ and $d + 1 > b + c$, i.e. $d \geq b + c \geq 5$. Analogously $e + 1 \geq c + d$, hence $e \geq c + d \geq 8$. And $f \geq d + e \geq 13$, $g \geq e + f \geq 21$ and $h \geq e + f \geq 34$. 1, 2, 3, 5, 8, 13, 21 and 34 satisfy the condition.

30. 2021 balls are arranged in a row and are numbered from 1 to 2021. Each ball is coloured in one of four colours: green, red, yellow or blue. Among any five consecutive balls there is exactly one red, one yellow and one blue ball. After any red ball the next ball is yellow. The balls numbered 2, 20 and 202 are green. What colour is the ball numbered 2021?

- (A) Green (B) Red (C) Yellow (D) Blue
(E) It is impossible to determine.

SOLUTION: Among any five consecutive balls there are one red, one yellow, one blue and two green. Consider any six consecutive balls. The mentioned above is true for the first five and for the last five balls. It follows that the first and the sixth balls have the same color. Hence, the color of any ball is repeated after five consecutive balls. Thus, the ball nr. 20 has the same color as the ball nr. 5 (since $20 = 5 \times 3 + 5$). The ball nr. 202 has the same color as the ball nr. 2 (since $202 = 5 \times 40 + 2$). Thus, the balls nr. 2 and nr. 5 are green. The ball nr. 2021 has the same color as the ball nr. 1 (since $2021 = 5 \times 404 + 1$). It follows that the ball nr. 3 is red, nr. 4 is yellow and nr. 1 is blue. Hence, the ball nr. 2021 is blue.